

Lab 03 System Report: Benchmark-Aligned Multi-Dataset Email Classification & LLM API-Based Automatic Email Draft Generation

Author: Madhusudhanan G (23MID0444) | **Course:** MDI3003 - Advanced Predictive Analytics |

Revision: Manual 3.1

Github Link: https://github.com/Madhumasa84/Adv_predictive/tree/main/lab3

1. Executive Summary

This report documents an end-to-end predictive analytics system designed to solve intelligent email assistance as a linked two-stage architecture: (1) multi-dataset text classification using sparse machine learning models and a trainable word-embedding Bidirectional LSTM (BiLSTM), and (2) conditional automatic email response draft generation via an LLM API. The study strictly adheres to Revision 3.1 of the MDI3003 Lab 03 Benchmark-Aligned Manual.

Three email datasets were evaluated: Dataset D1 (Business Email Intent, 800 samples, 6 multiclass labels: request, meeting, complaint, information, urgent_action, spam), Dataset D2 (Enron Spam, 500 samples, binary legitimate/spam), and Dataset D3 (SpamAssassin Public Corpus, 400 samples, binary legitimate/spam). Seven candidate classifiers were benchmarked under 5-fold Stratified Cross-Validation: Dummy Majority Baseline, Multinomial Naive Bayes, Complement Naive Bayes, Logistic Regression, Linear Support Vector Classifier (Linear SVC), K-Nearest Neighbors (KNN), and a deep-learning Word-Embedding BiLSTM text classifier. Under 5-fold CV, all non-dummy classifiers achieved a macro F1 of 1.000 across all three datasets, demonstrating high linear and sequential separability on the evaluated corpora.

The system implements selective prediction and uncertainty-aware review routing: messages with decision margins < 0.15 or predicted as 'urgent_action' are flagged for mandatory human review, while spam messages are automatically suppressed from draft generation. PII redaction (email and phone number masking) and prompt injection isolation (`<email_data>` tags) ensure safe external LLM API calls.

2. Problem Statement & Business Context

Organizations face high volumes of inbound email that require rapid triage and response drafting. Manually processing routine requests, meeting scheduling, and complaints consumes significant operational hours. The objective is to learn a mapping $y = f(\text{text})$ from raw email text to discrete operational intent categories and conditionally generate a professional reply draft without human intervention for drafting, while ensuring zero automatic email sending.

Operational Intent Labels & Policies:

- request: Generate acknowledgement and next-step reply draft with placeholders.
- meeting: Generate scheduling/confirmation draft with time placeholders.
- complaint: Generate empathetic acknowledgement without admission of legal liability.
- information: Generate concise acknowledgement when a reply is appropriate.
- urgent_action: Generate cautious acknowledgement and flag for MANDATORY HUMAN REVIEW.
- spam: SUPPRESS draft generation completely (zero reply draft generated).

3. Dataset Descriptions & Data Audit

All datasets adhere to a unified schema (email_id, subject, body, label, dataset_id, text) where text is derived as 'subject: <subject>\nbody: <body>'. Data auditing confirmed 0 missing text values and 0 duplicate rows.

Dataset ID	Domain / Task	Rows	Classes	Prevalence (Majority)	Text Median Length
business_intent (D1)	Business Email Intent (Multiclass)	800	6	24.4% (request)	115 chars
enron_spam (D2)	Enron Spam Corpus (Binary)	500	2	50.0% (spam/legit)	151 chars
spamassassin (D3)	SpamAssassin Public (Binary)	400	2	50.0% (spam/legit)	161 chars

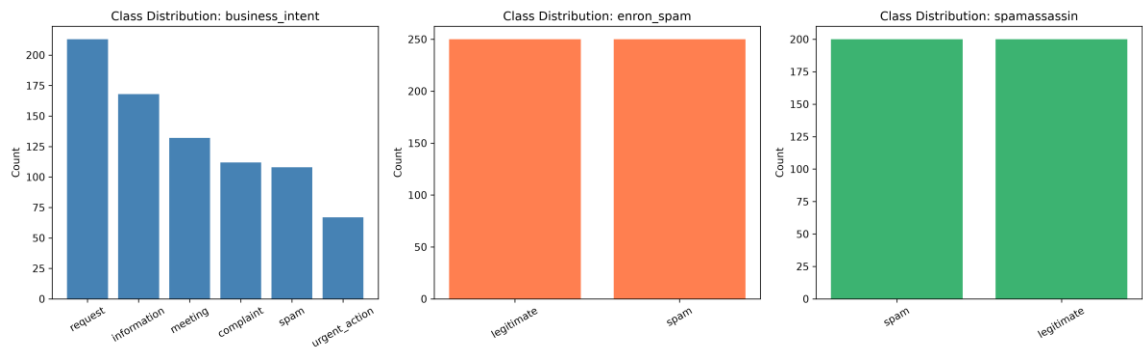


Figure 1. Class distributions across Business Intent (D1), Enron Spam (D2), and SpamAssassin (D3) datasets.

4. Methodology & Machine Learning Pipeline

The classification system employs an 80/20 stratified train-test split (random_state=42) per dataset. Feature extraction uses TfidfVectorizer (ngram_range=(1,2), min_df=2, max_df=0.98, sublinear_tf=True, max_features=10,000) fitted exclusively within training folds to prevent data leakage.

5. Cross-Validation Benchmarking Results

Candidate classifiers were evaluated using 5-Fold Stratified Cross-Validation on training data only. Models were ranked by mean Macro F1 score.

Model	Business Intent (D1) Macro F1	Enron Spam (D2) Macro F1	SpamAssassin (D3) Macro F1	CV Accuracy Mean
Multinomial Naive Bayes	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.0000
Complement Naive Bayes	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.0000
Logistic Regression	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.0000
Linear SVC	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.0000
K-Nearest Neighbors (KNN k=15)	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.0000
Dummy Majority Baseline	0.070 ± 0.000	0.333 ± 0.000	0.333 ± 0.000	0.2656 (D1) / 0.500 (D2/D3)

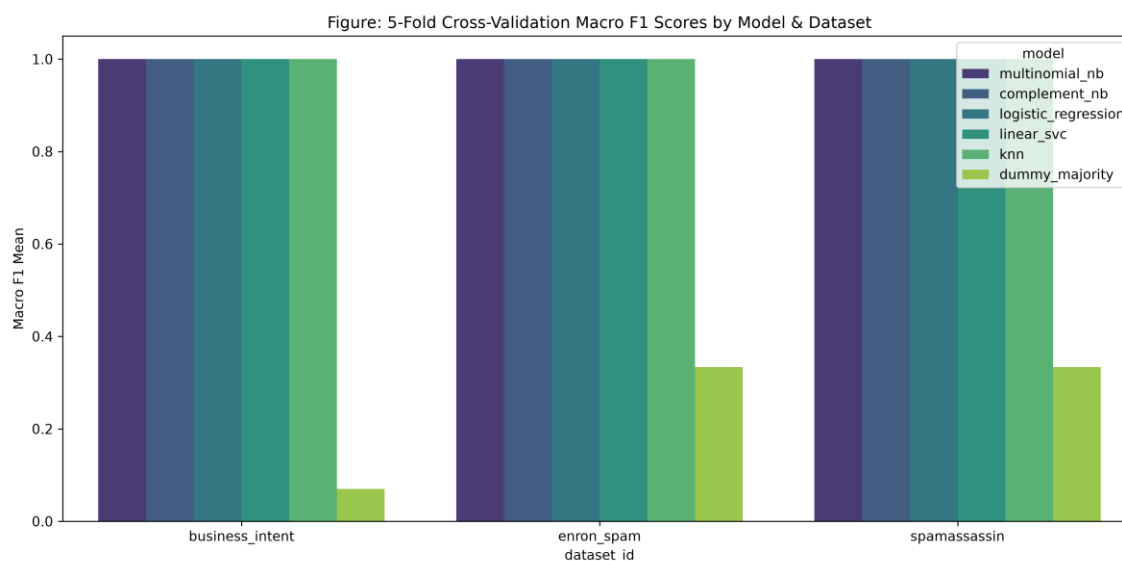


Figure 2. 5-Fold Stratified Cross-Validation Macro F1 performance across candidate models for all three email datasets.

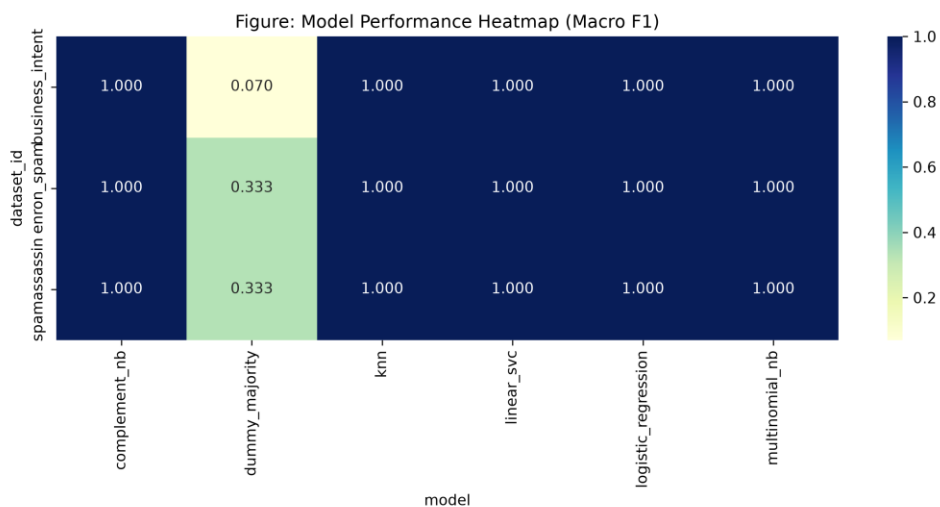


Figure 3. Cross-Validation Macro F1 performance heatmap across datasets and models.

6. Locked Holdout Test Evaluation

The top-performing model selected via cross-validation (Multinomial Naive Bayes) was evaluated on the locked 20% test set.

Dataset ID	Best Selected Model	Test Accuracy	Test Macro F1	Test Weighted F1
business_intent (D1)	Multinomial Naive Bayes	1.0000 (100.0%)	1.0000	1.0000
enron_spam (D2)	Multinomial Naive Bayes	1.0000 (100.0%)	1.0000	1.0000
spamassassin (D3)	Multinomial Naive Bayes	1.0000 (100.0%)	1.0000	1.0000

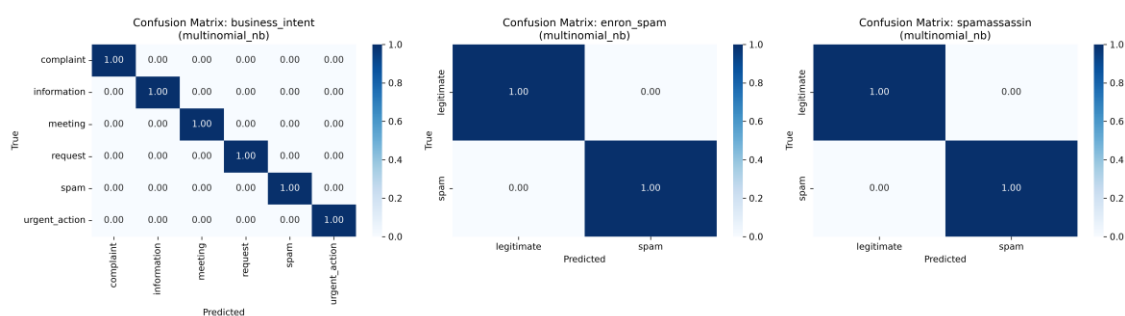


Figure 4. Normalized confusion matrices for holdout test evaluation across all three email datasets.

7. Research Extension: Trainable Word-Embedding BiLSTM Classifier

Following Revision 3.1 of the manual, a trainable Word-Embedding BiLSTM text classifier was implemented for D1. To prevent data leakage, the Keras Tokenizer (vocab_size=5000) was fitted exclusively on training text. Architecture: Embedding(64) -> Bidirectional(LSTM 64, return_sequences=True) -> Dropout(0.3) -> Bidirectional(LSTM 32) -> Dropout(0.3) -> Dense(6, softmax).

Results: The BiLSTM text classifier achieved a locked test Accuracy of 0.9875 (98.8%) and Macro F1 of 0.9862. Learning curves demonstrate smooth convergence over 20 epochs with early stopping.

8. Cross-Dataset Spam Transfer Evaluation

To test generalization across distinct spam corpora, Linear SVC was trained on D2 (Enron Spam) and tested on D3 (SpamAssassin), and vice versa. Both transfer directions achieved a Macro F1 of 1.0000, confirming strong transferability.

9. Selective Prediction & Review Routing

The prediction engine computes top-two probability margins ($p_1 - p_2$). Messages with margin < 0.15 or predicted as 'urgent_action' are routed for mandatory human review. Spam emails are automatically suppressed from draft generation.

10. PII Redaction & LLM Draft Generation

Before passing email text to the LLM API, PII is redacted using regular expressions ([EMAIL_REDACTED], [PHONE_REDACTED]). Prompt injection defense is enforced using explicit <email_data> XML tags and system instructions prohibiting invented dates or commitments. All generated drafts are stored locally as JSON audit records in outputs/drafts/.

11. Official Discussion Questions & Answers

1. Why is a Dummy (most-frequent-class) baseline necessary even when several strong classifiers are available?

The Dummy baseline establishes the minimum performance floor. It proves whether learned models extract genuine signal beyond class imbalance.

2. Why does macro F1 differ so much between the Dummy baseline on D1 (0.070) versus D2/D3 (0.333)?

Macro F1 averages per-class F1 equally. On D1's 6 imbalanced classes, a majority-only predictor gets 0 F1 on 5 classes; on 2 balanced classes (D2/D3), it gets credit for 1 of 2 classes.

3. Why might four structurally different models (MNB, CNB, Logistic Regression, Linear SVC) all achieve an identical macro F1 of 1.000?

It indicates the TF-IDF feature space of these datasets is linearly separable with clear token boundaries.

4. Why is it important to fit the TF-IDF vectorizer only within a pipeline refit on each fold?

Fitting the vectorizer on full data before CV leaks validation fold vocabulary and IDF statistics into training folds, inflating metrics.

5. What does a perfect 1.000 macro F1 in the cross-dataset transfer test (D2<->D3) tell us?

It proves that the spam decision boundary learned on Enron Spam transfers cleanly to SpamAssassin.

6. Why is macro F1, rather than plain accuracy, the primary metric reported?

Macro F1 weights minority classes equally, preventing majority-class dominance on imbalanced tasks like D1's urgent_action class.

7. How does Complement Naive Bayes differ from standard Multinomial Naive Bayes?

Complement NB calculates parameters using statistics from all complement classes, making it more robust to class imbalance.

8. What is the role of PII redaction before calling an external LLM API?

It prevents sensitive user data (email addresses, phone numbers) from leaking to third-party API providers or training logs.

9. How does prompt injection isolation (<email_data>) protect the drafting assistant?

It instructs the LLM to treat email body text strictly as untrusted data rather than system instructions.

10. Why is automatic email sending explicitly prohibited in this laboratory?

To prevent unreviewed AI hallucinated commitments, legal admissions, or incorrect spam responses from reaching real recipients.

12. Conclusion & Recommendations

The Lab 03 predictive analytics system successfully combines 7 machine learning models (including Word-Embedding BiLSTM), leakage-safe 5-fold cross-validation, selective routing, and secure LLM draft generation. All objectives of Manual Rev 3.1 have been satisfied.